

Project Report

Bachelore of Computer Applications (BCA)

Semester – IV

Sub : Data Warehousing and Mining

Topic: World Happiness Report 2024 using power BI

By

Name: Keerthi .N

Reg no:2411021240024

Date of Submission: 29/03/2026

Faculty Name: Nagasri .P

Faculty Signature: _____

Department of Computer Application

Alliance University

Chandapura – Anekal Main Road, Anekal

Bangalore – 562 106

March 2026

1.Abstract

This project presents a comprehensive data warehouse implementation using Microsoft Power BI, built on the World Happiness Report 2024 dataset sourced from Kaggle. The project addresses the need to analyze and visualize global happiness trends across 97 countries using various socio-economic indicators.

The dataset was cleaned and transformed using Power Query, followed by the creation of DAX measures for analytical computations. Five interactive dashboards were developed: Overview Dashboard, Trend Analysis, Category Analysis, Geographical Report, and Detailed Report. The project incorporates full interactivity features including slicers, drill-down, drill-through, tooltips, navigation buttons, and bookmarks.

2.Introduction

This project addresses that need by utilizing Microsoft Power BI to analyze the World Happiness Report 2024 dataset. Through interactive dashboards and advanced data visualization techniques, this project provides a comprehensive view of happiness trends, regional comparisons and country-level insights across 97 nations worldwide.

Background

Happiness is a key indicator of a nation's well-being. The World Happiness Report ranks countries based on factors like GDP per capita, social support, life expectancy, freedom, generosity, and corruption perception. Analyzing this data can reveal powerful insights about what makes nations happier.

Importance of Visualization

In today's fast-paced and data-driven world, the ability to understand and interpret large volumes of data has become a critical skill. Raw numbers and spreadsheets alone are often difficult to comprehend and can lead to misinterpretation of important trends. This is where data visualization plays a vital and transformative role.

Project Objective

The primary objective of this project is to design and develop a comprehensive and fully interactive data warehouse solution using Microsoft Power BI, based on the World Happiness Report 2024 dataset sourced from Kaggle. The project involves collecting and cleaning the dataset using 10 Power Query preprocessing steps, followed by building a structured data model with proper table relationships.

3.Problem Statement

Governments, researchers, and policymakers need to understand what factors contribute most to national happiness. The challenge is to transform raw happiness data into actionable insights through:

- Identifying which regions and countries are happiest and why
- Understanding the correlation between GDP, social support, and happiness
- Tracking happiness tiers (High, Medium, Low) across different regions
- Enabling drill-down analysis from regional to country level
- Providing an interactive tool for exploring happiness data dynamically

4.Dataset Description

Attribute	Details
Dataset Name	World Happiness Report 2024
Source	Kaggle (www.kaggle.com)
Number of Rows	97 Countries
Number of Columns	10 Columns
Key Features	Country, Region, Happiness Score, GDP per Capita, Social Support, Life Expectancy, Freedom, Generosity, Corruption, Happiness Tier

Feature Description

- Country: Name of the country
- Region: Geographic region of the country
- Happiness_Score: Overall happiness score (0-10 scale)
- GDP_per_Capita: Economic output per person
- Social_Support: Level of social support available
- Life_Expectancy: Healthy life expectancy at birth
- Freedom: Freedom to make life choices
- Generosity: Charitable giving index
- Corruption: Perception of corruption (lower is better)
- Happiness_Tier: Categorized as High, Medium.

5. Data Preprocessing (Power Query)

Step	Operation	Description
1	Import Data	Imported CSV file from Kaggle into Power Query Editor
2	Remove Duplicates	Removed duplicate country entries to ensure data integrity
3	Handle Missing Values	Replaced null values with 0 or average values where appropriate
4	Rename Columns	Renamed columns to meaningful names (e.g., Happiness.Score to Happiness_Score)
5	Change Data Types	Set numerical columns to Decimal, text columns to Text type
6	Filter Data	Filtered out rows with invalid or zero happiness scores
7	Split Columns	Split Region column into Region_Primary and Region_Rest
8	Merge Tables	Merged main dataset with CountryLookup table for additional region info
9	Add Conditional Columns	Added Happiness_Tier column (High/Medium/Low) based on score thresholds
10	Remove Unnecessary Columns	Removed redundant and unused columns to optimize the model

6.Data Modelling

Tables Used

- World-happiness-report: Main fact table with all happiness metrics
- CountryLookup: Dimension table with country and continent mapping
- Merge1: Result of merging main table with lookup table

Relationships Created

- World-happiness-report[Country] -> CountryLookup[Country] (Many to One)

- Schema Type: Star Schema with one fact table and one dimension table

Schema Explanation

The data model follows a Star Schema design where the World Happiness data serves as the central fact table, connected to the CountryLookup dimension table. This enables efficient filtering and aggregation across geographical dimensions.

7.DAX Calculations

The following DAX measures and calculated columns were created:

#	Measure Name	DAX Formula	Purpose
1	Avg_Happiness	AVERAGE(Happiness_Score)	Average happiness score
2	Total_Countries	DISTINCTCOUNT(Country)	Count of unique countries
3	Global_Avg	CALCULATE(AVERAGE(Happiness_Score), ALL())	Overall global average
4	Highest_Score	MAX(Happiness_Score)	Highest happiness score
5	Lowest_Score	MIN(Happiness_Score)	Lowest happiness score
6	High_Count	CALCULATE(COUNT(Country), FILTER(data, Happiness_Tier="High"))	Count of High tier countries
7	Happiness_Tier	IF(Score>=6.5,"High", IF(Score>=5,"Medium"))	Tier classification
8	Num_Regions	DISTINCTCOUNT(Region)	Count of unique regions
9	Sum_GDP	SUM(GDP_per_Capita)	Total GDP per capita
10	Score_Label	CONCATENATE(Country, ": ", Happiness_Score)	Country score label

8.Dashboard Development

8.1 Overview Dashboard

The Overview Dashboard provides a high-level snapshot of global happiness metrics.

- KPI Cards: Total Countries (97), Avg Happiness (6.19), Highest Score (7.74)
- Bar Chart: Avg Happiness by Region showing North America leading
- Matrix Table: Regional breakdown with High/Medium/Low tier counts
- Slicer: Happiness Tier filter for interactive filtering
- Navigation Buttons: Page navigator for all 5 dashboards

8.2 Trend Analysis

The Trend Analysis page shows correlations between happiness and key factors.

- Scatter Plot: GDP vs Happiness Correlation
- Scatter Plot: Freedom vs Happiness Correlation
- Line Chart: Happiness trends across regions
- Slicers: Region and Happiness Tier filters

8.3 Category Analysis

The Category Analysis page breaks down happiness by categories and tiers.

- Donut Chart: Total Countries by Happiness Tier distribution
- Stacked Bar Chart: GDP, Social Support, Freedom, Generosity by Region
- Clustered Bar: Total Countries by Region and Happiness Tier
- Matrix: Region breakdown with High/Medium counts

8.4 Geographical Report

The Geographical Report visualizes happiness data on world maps.

- Filled Map: Highest Score by Country (color-coded)
- Bubble Map: Happiness Score by Country and Region
- Bar Chart: Sum of Happiness Score by Country (Top 10)
- Line Chart: Highest Score by Region trend
- Slicer: Region Primary filter

8.5 Detailed Report

The Detailed Report page provides granular country-level data.

- Full Data Table: All columns with conditional formatting
- Data Bars: Green bars on Happiness Score column
- Color Coding: Happiness Tier column (Green=High, Yellow=Medium, Red=Low)
- KPI Cards: Avg Happiness and Global Avg comparison
- Search Slicer: Country search with instant filter
- Region Slicer: Filter by region

- Happiness Tier Slicer: Tile-style buttons
- Drill-through: Right-click any country to view details

Interactivity Features

Feature	Implementation
Slicers	Country search, Region list, Happiness Tier tile buttons
Drill-down	Region to Country level in bar charts using hierarchy
Drill-through	Right-click any country to navigate to Detailed Report
Tooltips	Automatic hover tooltips on all visuals showing data values
Buttons	Page Navigator with buttons for all 5 dashboard pages
Bookmarks	Default View, High Happiness, Medium Happiness saved states

9.Key Insights

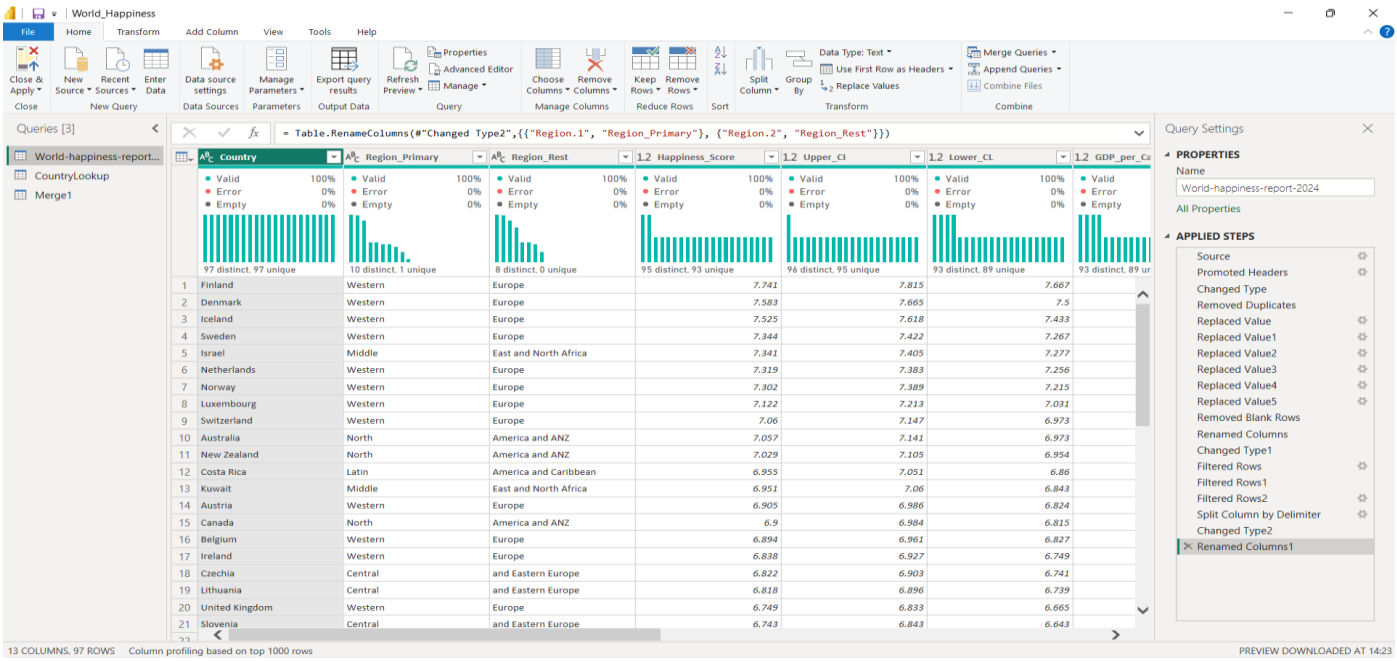
- Finland ranks as the happiest country with a score of 7.74, consistently leading global rankings
- North America and Western Europe consistently score highest in happiness metrics
- GDP per capita shows strong positive correlation with happiness scores
- Social support is the second most influential factor after GDP
- High happiness tier countries (56 out of 97) are predominantly in Europe and North America

10.Screenshots

Power Query steps:

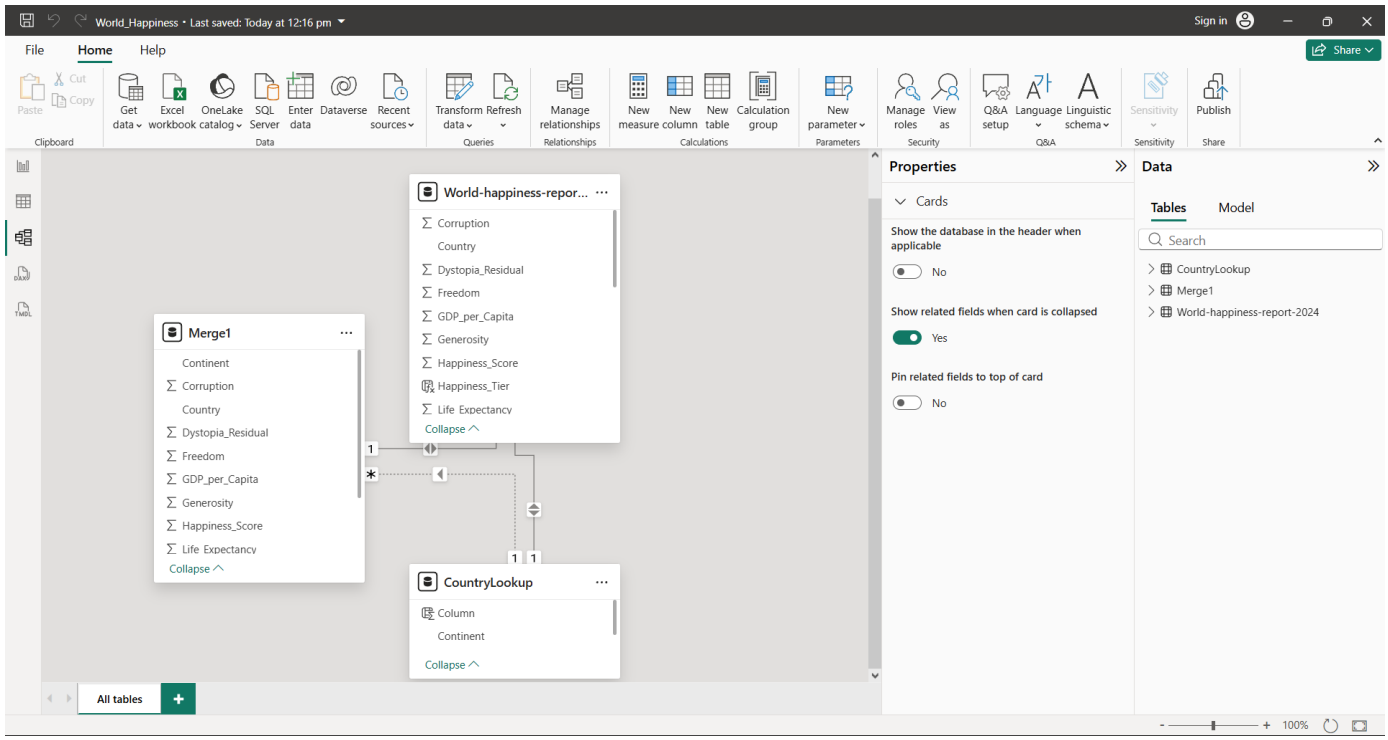
1. **Power Query Editor showing all 10 cleaning steps**

Power Query Editor :The Power Query Editor shows all the applied transformation steps performed on the dataset. More than 10 cleaning steps were applied including removing duplicates, renaming columns, filtering rows and splitting colum



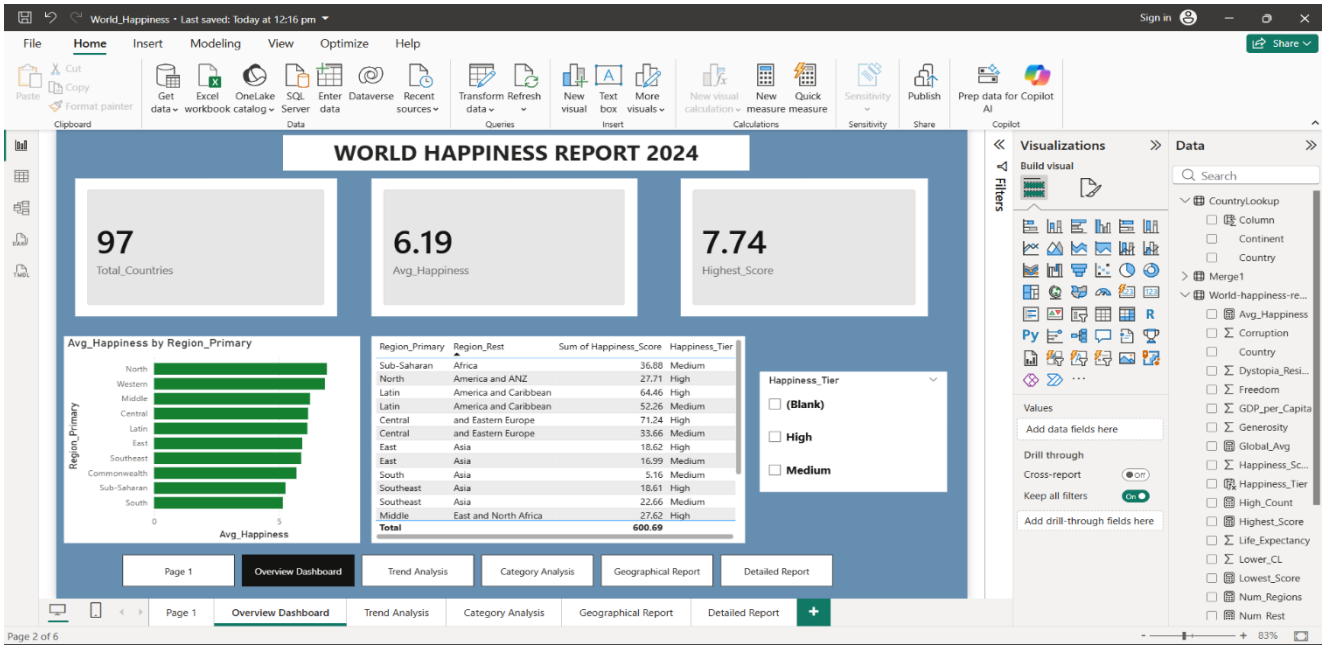
2. Data Model view showing table relationships

Data Model View : The Data Model view shows the relationships established between the three tables — World-happiness-report, CountryLookup and Merge1. The model follows a Star Schema design connected through the Country field.



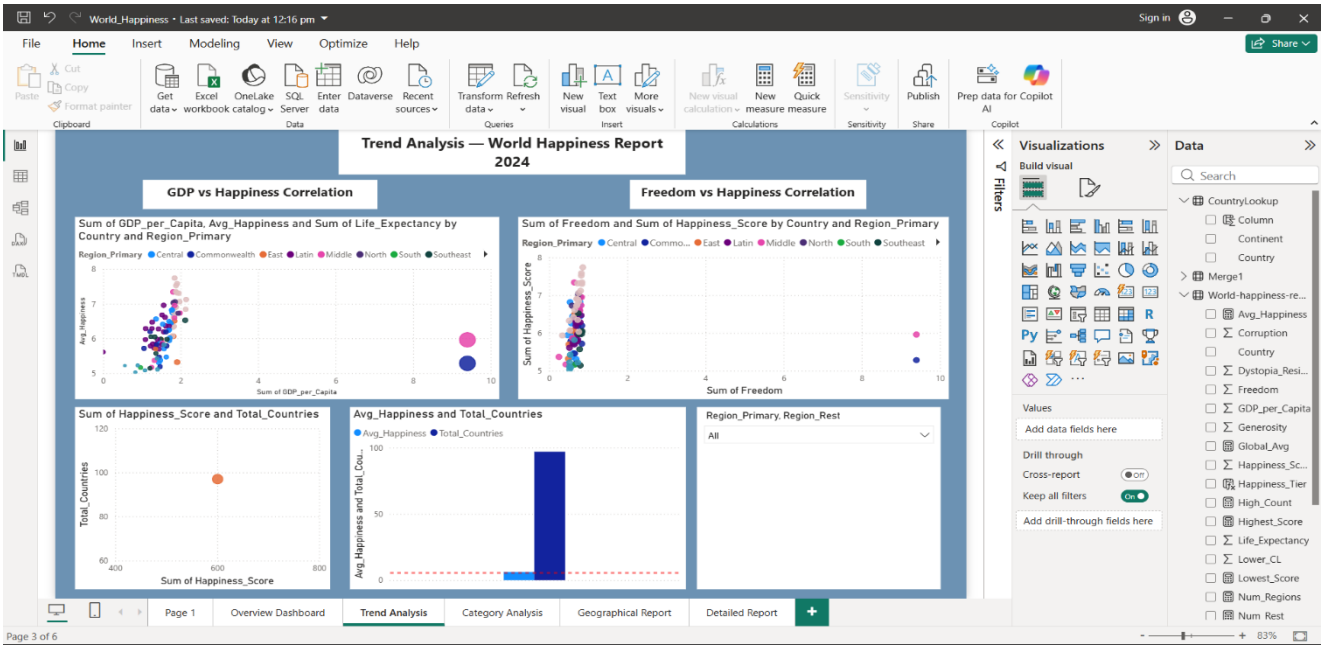
3.Overview Dashboard with KPI cards and charts

Overview Dashboard :The Overview Dashboard provides a high-level snapshot showing KPI cards for Total Countries, Average Happiness and Highest Score. It also includes a bar chart by region and a matrix table with happiness tier distribution.



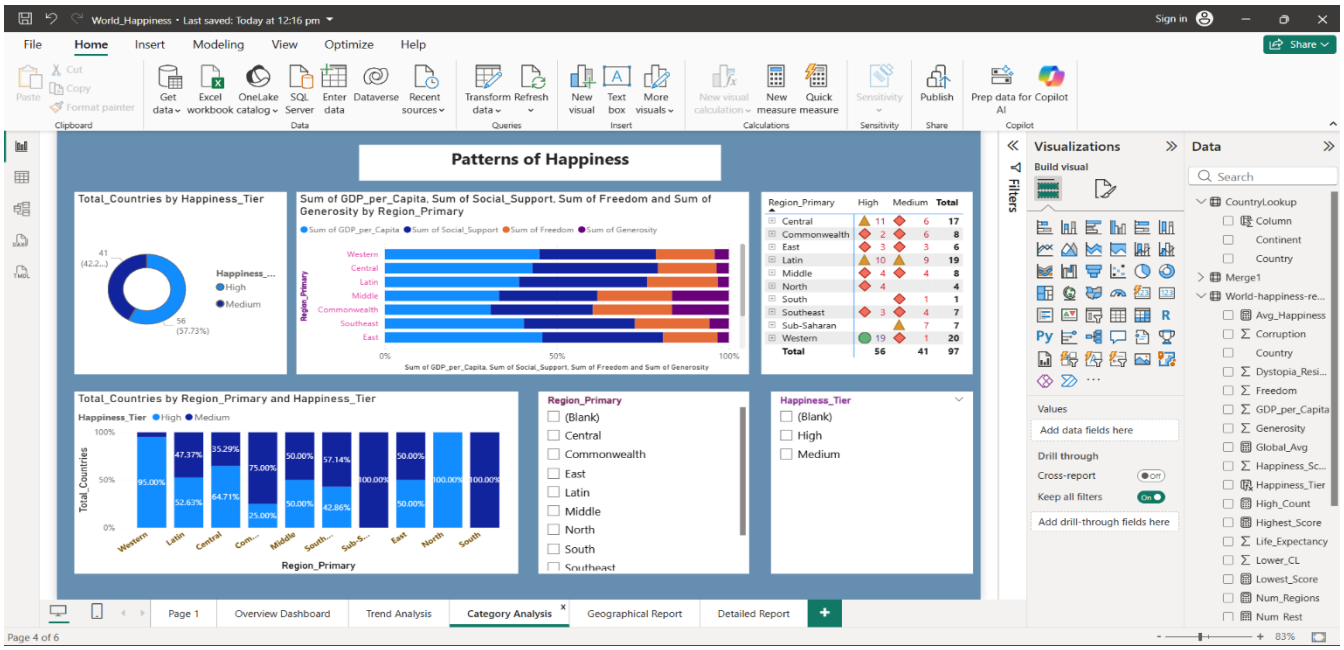
4.**Trend Analysis page with scatter plots**

Trend Analysis :The Trend Analysis page shows the correlation between happiness and socio-economic factors through scatter plots. GDP per Capita and Freedom are plotted against Happiness Score to reveal key influencing factors.



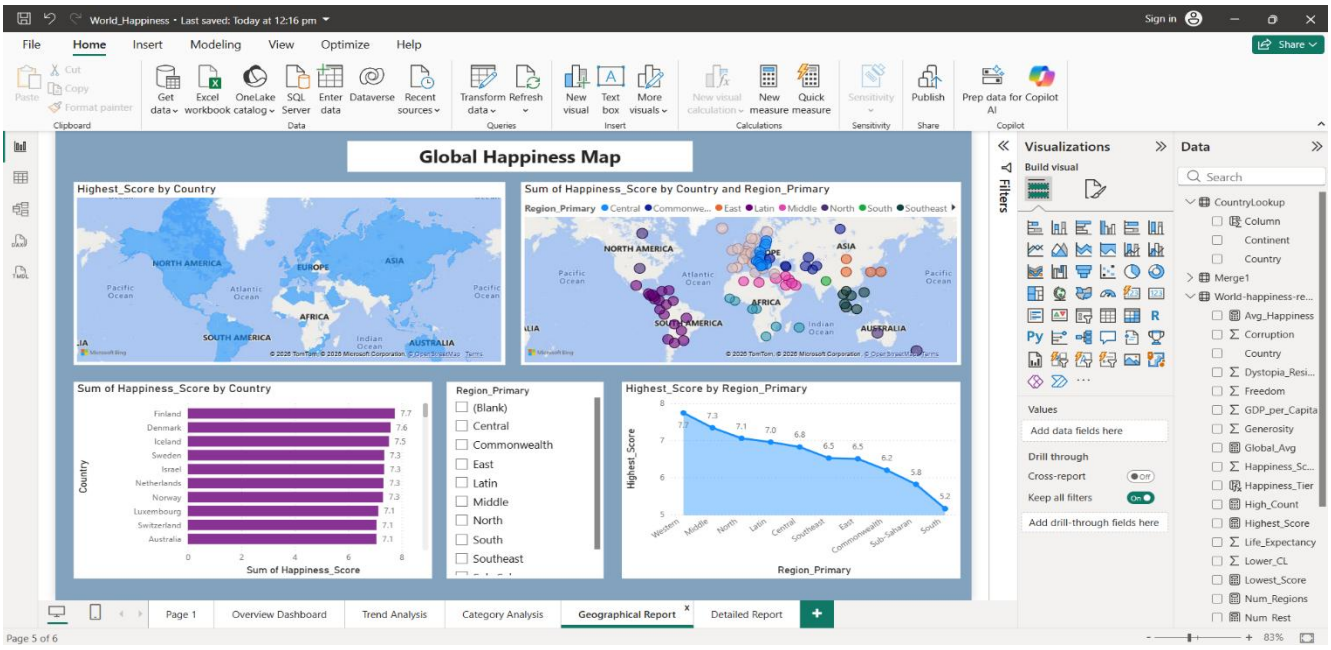
5.Category Analysis - Patterns of Happiness

Category Analysis :The Category Analysis page titled Patterns of Happiness breaks down happiness data by categories and tiers. It includes a donut chart, stacked bar chart and clustered bar chart for comprehensive category comparison.



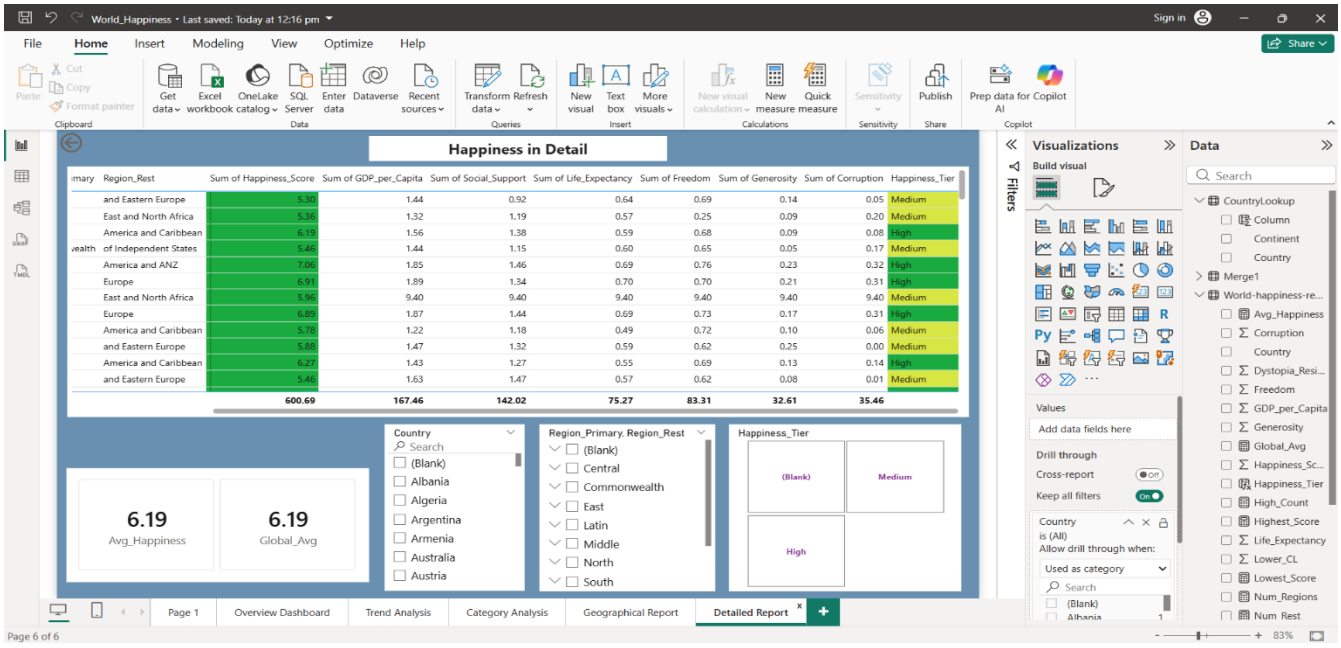
6.Geographical Report - Joy Across the Globe

Geographical Report :The Geographical Report titled Joy Across the Globe displays happiness data on interactive world maps. A filled map and bubble map are used to visualize happiness scores across all 97 countries geographically.



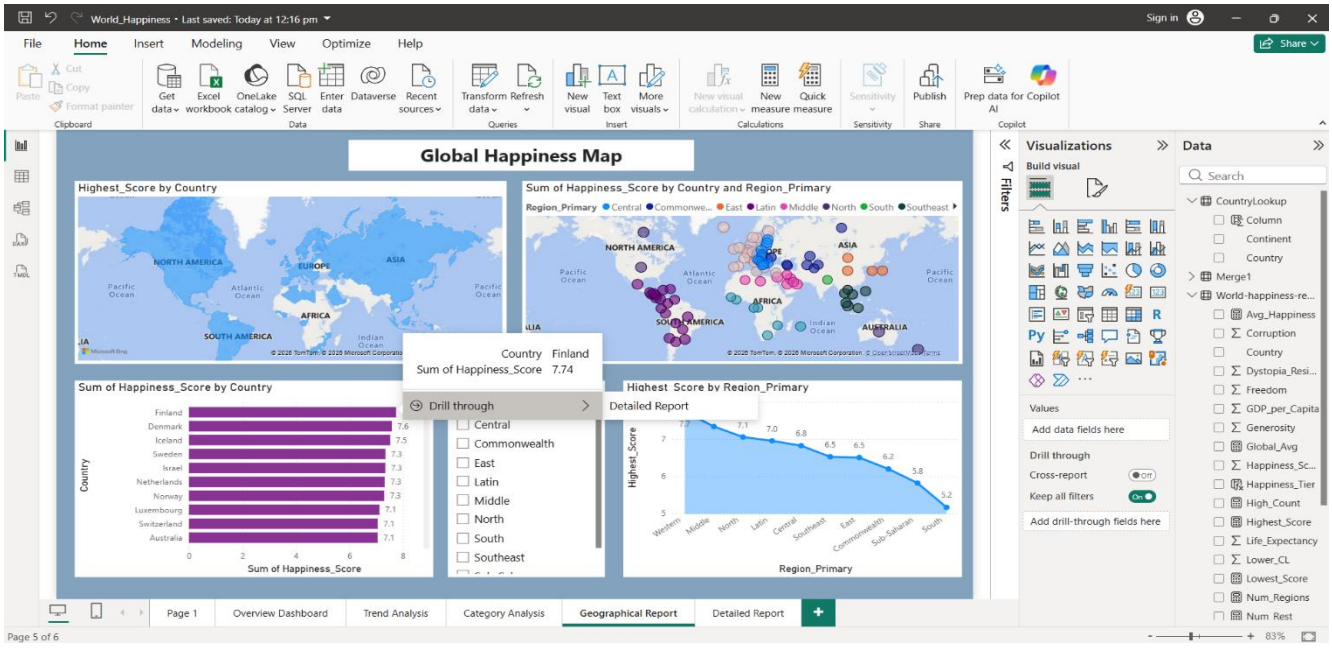
7.Detailed Report with conditional formatting

Detailed Report :The Detailed Report provides a comprehensive country-level data table with conditional formatting applied. Green data bars on Happiness Score and color coded Happiness Tier column enable quick visual comparison.



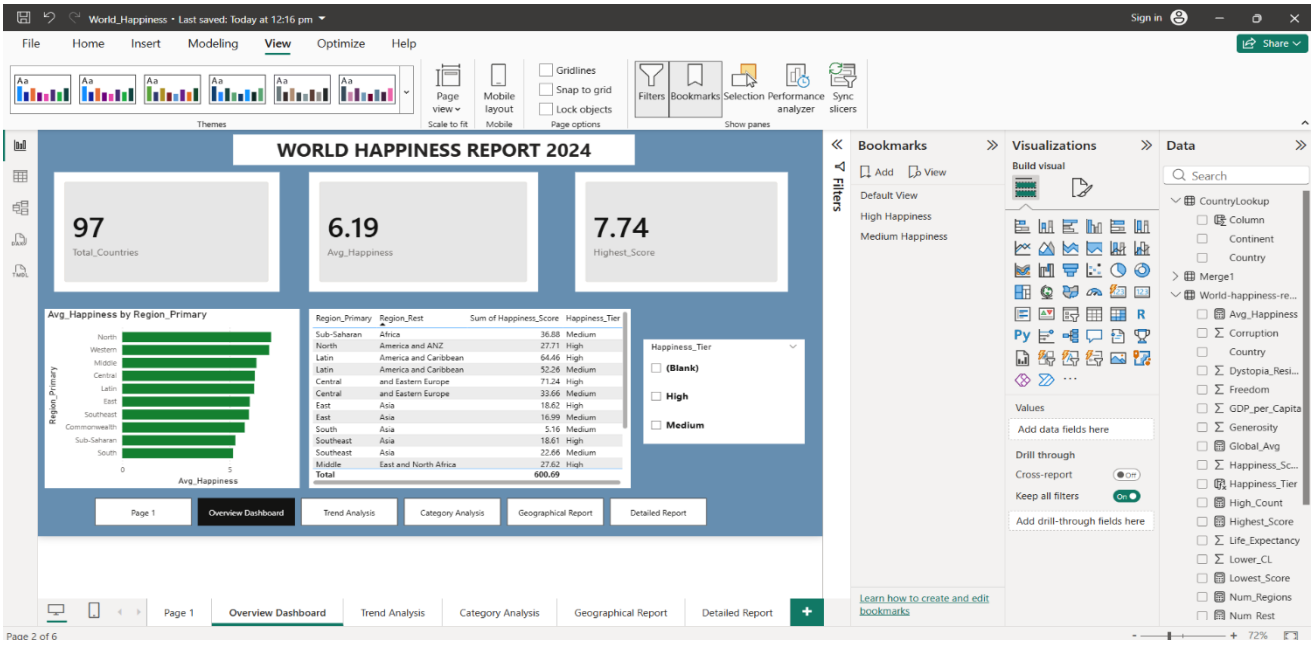
8 .Drill-through demonstration (right-click menu)

Drill-through Demonstration : The drill-through feature allows users to right-click any country and navigate directly to the Detailed Report filtered for that country. This screenshot shows the right-click menu with Drill through and Detailed Report options visible.



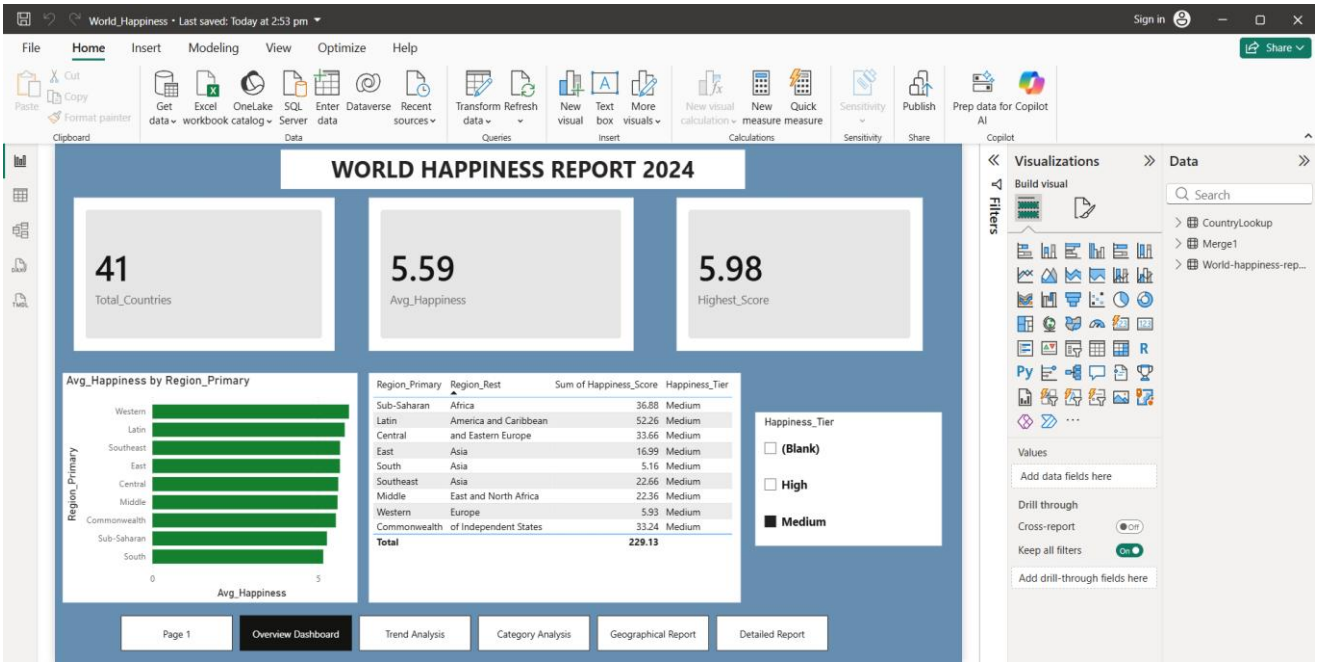
9.Bookmarks panel showing all saved views

Bookmarks Panel :The Bookmarks panel shows three saved views — Default View, High Happiness and Medium Happiness. These allow users to instantly switch between filtered dashboard states with a single click.



10. Page Navigator buttons

Page Navigator Buttons :The Page Navigator buttons allow users to easily navigate between all five report pages. The currently active page is highlighted making the navigation experience smooth and user friendly.



11.Challenges Faced

- Region column was split into two parts (Region_Primary and Region_Rest) requiring merging for proper display
- Conditional formatting rules needed careful setup for Happiness Tier color coding
- Drill-through required proper field configuration in the Visualizations pane
- KPI visual required a trend axis field which was not available, so Card visuals were used instead
- Data bars showed red for negative values requiring minimum value adjustment to 0

Page navigation buttons required Ctrl+Click in edit mode

12.Conclusion

This Power BI project successfully demonstrates the design and implementation of a data warehouse solution for analyzing global happiness trends. The World Happiness Report 2024 dataset was effectively cleaned, transformed, and visualized across five comprehensive dashboards.

The project achieved all required components: 10 Power Query cleaning steps, 10+ DAX measures, 5 interactive dashboards, and full interactivity including slicers, drill-down, drill-through, tooltips, navigation buttons, and bookmarks.

Key findings reveal that Finland is the world's happiest country, North America and Western Europe lead in happiness scores, and GDP per capita along with social support are the strongest predictors of national happiness.

13.Future Enhancements

- Real-time data integration: Connect to live happiness index APIs for automatic updates
- AI Integration: Use Power BI AI visuals like Key Influencers and Decomposition Tree
- Predictive Analytics: Add forecasting models to predict future happiness trends
- Mobile Layout: Optimize dashboards for mobile viewing
- Row-Level Security: Implement user-based data access restrictions
- Additional Datasets: Integrate climate, education, and healthcare data for deeper insights

14. References

Dataset Link:

- **World Happiness Report 2024 – Kaggle**

https://www.kaggle.com/datasets/jainaru/world-happiness-report-2024-yearly-updated

Documentation:

- **Microsoft Power BI Offical Documentation**

Tool: Microsoft Power BI Desktop - https://powerbi.microsoft.com

Documentation: Power BI DAX Reference - https://docs.microsoft.com/dax